Predicting Mental Health Treatment Needs Using Behavioral and Occupational Data

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***Abstract*—Mental health disorders are a growing global public health challenge affecting individuals across demographic and occupational groups. Early detection and intervention is hindered by limited access to timely diagnosis and the inefficiency of traditional healthcare systems in scaling up services. Further, the fact that many people are going undiagnosed is because of social stigma, unawareness, and limited access to healthcare. The purpose of this study is to create a scalable machine learning model for predicting mental health treatment needs based on behavioral, psychological, occupational, and demographic data. Apache Spark and Python were used to build an end-to-end pipeline to process 292,364 survey responses, including data cleaning, removal of duplicates, and encoding categorical features with OrdinalEncoder. The models were trained and optimized utilizing ensemble methods such as Random Forest and XGBoost with RandomizedSearchCV and stratified cross-validation. Experimental results demonstrate that XGBoost is the best model with a mean accuracy of about 81% and an AUC of 0.93. Family history of mental illness, mood swings, stress, and availability of care options were key factors that predicted it. The results show the efficacy of scalable machine learning systems in early identification of mental health risks and in data-driven decision making for healthcare systems.**

***Index Terms*—Mental Health, Machine Learning, XGBoost, Random Forest, Apache Spark, Predictive analytics, Healthcare Systems**

I. Background of the Study

A. Generic Information

Mental health disorders have emerged as a major global public health challenge, affecting individuals across diverse demographic, occupational, and socio-economic groups [13]. Conditions such as stress, anxiety, and depression are increasingly prevalent due to modern lifestyle pressures, academic and work-related stress, and reduced social interaction [4]. Traditional mental health assessment methods rely heavily on clinical evaluations and self-reported questionnaires, which are often time-consuming, subjective, and difficult to scale across large populations [11], [12]. As a result, many individuals experiencing mental health issues remain undiagnosed or receive delayed treatment [5], [8]. With the rapid growth of digital health data and advancements in machine learning, there is increasing potential to analyze behavioral and psychological patterns to support early and scalable detection of mental health risks [16], [9].

B. Problem Statement

People with mental health problems may not be identified due to their fear of being judged for their problems, lack of awareness, financial limitations or limited access to healthcare services [10]. Therefore, early help is frequently delayed and mental health issues can develop and become more severe over time [8]. In this project, an initial dataset containing 292,364 survey responses was analyzed. After preprocessing and duplicate removal, 97,723 unique responses were retained for model development. Of those responses, about 48.5% said they needed mental health treatment and 51.5% said they did not, creating a relatively balanced dataset for predictive modeling. The distribution of treatment needs in the data is shown in Figure 1. These findings highlight the importance of developing intelligent and scalable predictive systems capable of identifying individuals who may require mental health support at an early stage using behavioral and psychological patterns.

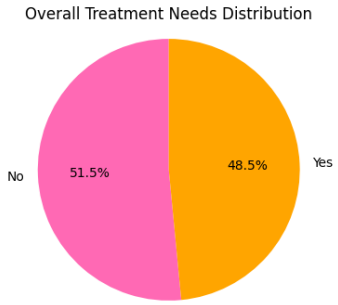


Fig. 1. Distribution of mental health treatment need in the dataset.

C. Aim and Objectives of the Work

This study aims to present a scalable machine learning framework for predicting mental health treatment needs based on behavioral, psychological, occupational, and demographic data derived from large-scale survey responses. The main objective is to create a predictive end-to-end solution that is able to process big data efficiently, find relevant patterns in complex behavioral attributes, and provide accurate classification. In this study, a key emphasis is placed on incorporating effective data preprocessing, feature engineering, and ensemble learning methods to improve the model's performance. In addition, it

seeks to systematically evaluate and compare several machine learning models, including Random Forest and XGBoost, to identify the best model to assist in the early identification of individuals in need of mental health care.

D. Contributions of the Work

The key contributions of this project are:

- Built a preprocessing pipeline using Apache Spark for large-scale survey data.
- Cleaned and processed 292,364 raw survey responses into 97,723 unique and machine learning-ready records.
- Performed exploratory data analysis to identify significant behavioral, demographic, and psychological trends associated with mental health care needs.
- Implemented Random Forest and XGBoost ensemble learning models using machine learning pipelines.
- Applied RandomizedSearchCV with stratified cross-validation for hyperparameter optimization and model improvement.
- Evaluated the performance of the model in terms of accuracy, precision, recall, F1-score, confusion matrix, and ROC AUC.
- Recognized factors that predicted the problem such as family history, care options, increasing stress, and mental health history.
- Achieved the best performance using XGBoost with approximately 81.4% accuracy and an AUC score of 0.93 while reducing false negative predictions.

E. Organization of the Report

The rest of this report is structured as follows. Section II presents the literature review related to mental health prediction, ensemble machine learning, and big data analytics in healthcare. Proposed methodology is detailed in Section III which outlines the data preprocessing steps, exploratory data analysis, feature encoding, model building, hyperparameter optimization, and model evaluation processes. The experimental results, model comparison, ROC-AUC analysis, confusion matrix evaluation, and feature importance findings are provided in Section IV. Finally, in Section V the results are summarized, limitations are discussed, and suggestions are offered for future research and practice.

II. Related Work

A. Machine Learning for Mental Health Prediction

Recent research has widely explored the use of machine learning models to predict mental health treatment needs using structured survey and behavioral datasets [1]. Research indicates that demographic, occupational and psychological characteristics can be efficiently categorized into groups of individuals likely to seek mental health services [14], [17]. Traditional models such as Logistic Regression and basic ensemble methods like Random Forest have been commonly applied in this domain. While they may have acceptable predictive accuracy, they often fall short in modeling complex, non-linear relationships within behavioral data and may be ill-suited to large-scale, real-world applications [2].

B. Ensemble Learning Approaches

Random Forest and Gradient Boosting models are among the ensemble methods that have proven to be highly successful in the context of tabular healthcare data. Research consistently demonstrates that boosting-based methods like XGBoost perform better than bagging-based methods in terms of prediction accuracy and handling feature interactions more effectively [3], [7]. Across multiple studies, ensemble models have been shown to improve classification performance in mental health prediction tasks by reducing variance and increasing generalization capability. In this project, this trend is also observed, where XGBoost achieves higher accuracy and better recall compared to Random Forest.

C. Key Predictive Factors and Feature Importance

Family history, mental health history, stress and behavioral changes are all well-established factors that predict mental health treatment needs. Factors such as mood instability, coping behavior, and social support structures are also frequently highlighted as important indicators. Similar results were found in this study with features like family_history, care_options, growing_stress and mood_swings being all very influential in prediction [6]. This consistency confirms that behavioral and psychological indicators play a critical role in mental health classification models.

D. Big Data and Scalable Mental Health Analytics

Recently, the subject of scalable data processing frameworks has gained significant attention, especially in the context of large-scale healthcare data [2]. With large amounts of mental health survey data, distributed computing systems like Apache Spark are becoming more popular for efficiently processing the data [11]. This enables quicker preprocessing, cleaning, and transformation of datasets with hundreds of thousands of records. In this study, Spark was used to process over 292,000 survey responses, demonstrating its effectiveness in managing real-world big data challenges in mental health analytics.

E. Model Performance and Evaluation Insights

Past research indicates that moderate accuracy for mental health prediction tasks is usually achieved with machine learning models like Random Forest, and better-performing models like XGBoost consistently perform better [15]. As seen in the literature, boosting methods perform better on recall and F1-scores and, therefore, are more appropriate for sensitive applications, such as healthcare prediction, where false negatives are of paramount importance [3]. Additionally, XGBoost was found to be more accurate and to produce a higher ROC-AUC in this study, further demonstrating its superior performance in distinguishing between mental health treatment and non-treatment compared to the Random Forest model.

F. Research Gap

While previous research has shown the potential of machine learning in mental health prediction, the majority of these studies suffer from limited datasets, less detailed preprocessing methods, or more limited modeling approaches. There are many works that only use traditional machine learning techniques, with no attempt to use scalable big data frameworks or to perform significant hyperparameter optimization.

This study overcomes these limitations by:

- Analyzing a massive dataset of 292,364 records using Apache Spark.
- Cleaning and reducing data to 97,723 high-quality instances.
- Utilizing advanced ensemble models (Random Forest and XGBoost).
- Performing systematic hyperparameter tuning with cross-validation.
- Using XGBoost to demonstrate better prediction accuracy.
- Reducing false negatives, which is critical in mental health prediction.

Overall, this research presents a more scalable, optimized, and reliable machine learning framework for predicting mental health treatment needs than existing approaches.

III. Methodology

This section describes the machine learning framework that was developed to predict mental health treatment needs using behavioral, psychological, occupational, and demographic survey data from many people. The entire process is structured into the following stages: Data acquisition, Big Data processing, Data preprocessing, Exploratory data analysis, Feature preparation, Model development, Model evaluation, Model deployment. The overall process is implemented as a modular pipeline ensuring the repeatability, scalability and robustness of the process as a solution to large-scale heterogeneous data.

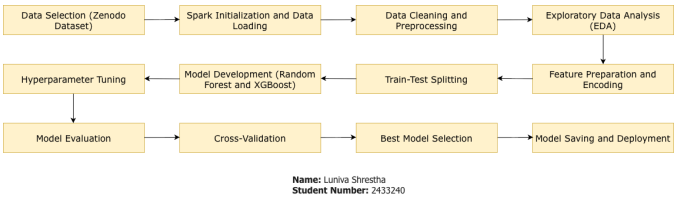


Fig. 2. Overall workflow of the proposed mental health prediction framework

Figure 2 shows the complete pipeline of the proposed mental health prediction framework. The workflow is broken down into a series of step-by-step steps for scalability and reproducibility. It covers data selection from Zenodo dataset, data loading and processing using Spark, data cleaning, exploratory data analysis (EDA), feature encoding and preparation, stratified train-test splitting, model development using Random Forest and XGBoost, hyperparameter tuning, model evaluation, cross validation, final model selection and saving the model for deployment.

A. Dataset Selection

The dataset for this study was downloaded from Zenodo [18] open access data repository, and includes 292,364 survey responses. These responses reflect behavioral, psychological factors, work-related and demographic attributes related to mental health treatment-seeking behavior. The dependent variable is treatment, defined as the presence or absence of mental health care-seeking (answered "Yes" or "No" to the question).

The dataset was chosen because it is large, has high variability in categorical features, and is highly relevant for mental health analytics in real-world settings. The flexibility of these properties allows it to be used for supervised machine learning problems with non-linear classification.

The key attributes are gender, occupation, family history of mental illness, stress levels, mood swings, period of indoor isolation, mental health history, social weakness, and the options of mental health care. These variables together are indicators of behavioral and environmental factors that can affect mental health outcomes.

B. Big Data Environment Setup

To efficiently process large-scale data, Apache Spark was utilized. PySpark was configured to enable distributed data processing and scalable computation across large datasets The experiments were executed in Google Colab using a high-RAM runtime environment with GPU acceleration support, ensuring efficient processing of large-scale data.

Spark was selected over traditional single-node tools such as Pandas due to its ability to handle distributed computation, in-memory processing, and fault-tolerant execution. Although Pandas was later used for modeling, Spark was essential during early-stage data ingestion and preprocessing to efficiently manage memory-intensive operations such as duplicate removal and schema inference on a large dataset.

The following tools and libraries were used:

- PySpark: distributed data processing for large-scale datasets
- Pandas and NumPy: data manipulation and numerical operations
- Matplotlib and Seaborn: exploratory visualization
- Scikit-learn: machine learning pipeline development
- XGBoost: gradient boosting classification

A Spark session was initialized using SparkSession.builder, enabling structured distributed data processing and efficient dataset loading in DataFrame format.

C. Data Loading and Inspection

The dataset was loaded from Google Drive into a Spark DataFrame using CSV format with automatic schema inference. Initial inspection was performed to understand the structure, data types, and quality of the dataset.

The dataset initially contained:

- 292,364 records
- 17 attributes

This stage enabled identification of missing values, duplicate entries, categorical distributions, and irrelevant features that required removal or transformation prior to modeling.

D. Data Preprocessing

Data preprocessing was done to enhance the data quality and make it compatible to machine learning algorithms.

1) Handling Missing Values: The missing values for the feature *self_employed* are filled with the label "Unknown". This method not only maintains the integrity of the data set, but avoids unnecessary data loss, particularly in large behavioral datasets where missingness can also contain implicit information.

```
Missing values filled with 'Unknown'
New shape: (292364, 17)
```

Fig. 3. Handling missing values in the self_employed column

2) Feature Removal: The Timestamp feature was removed as it does not contribute predictive value for classification tasks and may introduce noise or temporal bias unrelated to mental health outcomes.

3) Duplicate Removal: Duplicate records were eliminated using Spark's dropDuplicates() function to reduce redundancy and prevent bias from multiple survey responses, which might overestimate the model performance.

```
Number of duplicate rows found: 194641
Duplicates removed successfully
New shape: (97723, 16)
```

Fig. 4. Removal of duplicate records

After preprocessing, the dataset was reduced to:

- 97,723 records
- 16 features

4) Data Conversion: The Spark DataFrame was cleaned up and then transformed into a Pandas DataFrame for easy integration with the Scikit-learn and visualization libraries. This mixed model will provide flexibility in the model development process and scalability in preprocessing.

E. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was carried out to gain insights into the distributions of data and the relationships among the variables. Count plots, pie charts, and heatmaps were used for various visualization techniques.

Stress level, mood changes, family history, occupational category and the pattern of isolation in the home were considered key behavioral and psychological factors and were analyzed to see how they affect treatment seeking. This step gave some basic information for feature selection and model interpretation.

F. Feature Preparation and Encoding

Machine learning algorithms require numerical inputs; therefore, categorical variables were transformed into encoded representations.

1) Train-Test Split: The dataset was split into:

- 80% training set
- 20% testing set

```
Train-test split completed
X_train shape: (78178, 15)
X_test shape: (19545, 15)
y_train shape: (78178,)
y_test shape: (19545,)
```

Fig. 5. Train-test split of the dataset into training and testing sets

To ensure unbiased evaluation, a stratified sampling strategy was used to maintain the distribution of classes in training and testing sets.

2) Target Encoding: The categorical labels of the target variable *treatment* were transformed into binary numeric values using LabelEncoder for input into the classification algorithms.

3) Feature Encoding: A pipeline using OrdinalEncoder was used to encode categorical features. This way the transformation is applied the same way all of the time, training and testing. Unknown categories were handled using:

- handle_unknown = 'use_encoded_value'
- unknown_value = -1

4) Pipeline Construction: A machine learning pipeline was built for the integration of preprocessing and modeling in a single procedure. This will guarantee the reproducibility of the results, avoid data leakage and keep feature transformations consistent throughout the model development process.

G. Model Development

Two ensemble machine learning models were implemented due to their strong performance on structured tabular data and ability to capture complex non-linear relationships:

- Random Forest Classifier
- XGBoost Classifier

1) Random Forest: Random Forest is an ensemble learning algorithm that creates several decision trees and combines their predictions to achieve more stable predictions and lower variance in the results. To reduce overfitting and increase model diversity, each tree is trained on a bootstrap sample of the data using only a random subset of features.

The final prediction is made based on majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_N(x)\}$$

where $h_i(x)$ represents the prediction from the i-th decision tree and N is the total number of trees.

2) XGBoost: XGBoost (Extreme Gradient Boosting) is a modified version of the GBDT algorithm that sequentially creates and adds decision trees to correct the errors of the previous ones. It includes regularization to reduce the complexity of the model and improve generalization.

The objective function is:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

In the above equation, $l(y_i, \hat{y}_i)$ is a loss function that quantifies the prediction error, while $\Omega(f_k)$ is a regularization function for penalizing the complexity of the model. This formulation allows XGBoost to be highly predictive without overfitting.

3) Hyperparameter Tuning: The model's hyperparameters were efficiently optimized using RandomizedSearchCV across a large search space. This method is computationally more efficient than GridSearchCV while remaining equally effective.

The following parameters were tuned:

- Number of estimators
- Learning rate
- Maximum depth
- Subsampling ratio
- Minimum child weight

To ensure reliable performance estimation with balanced class distributions, stratified cross-validation was used during tuning.

H. Model Evaluation

Several different complementary metrics have been used to assess the model performance: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC score. These metrics collectively offer an overall assessment of the classification performance, especially in imbalanced and high risk prediction problems like that of mental health assessment.

I. Cross-Validation

Stratified K-Fold Cross-Validation (k=5) was used to evaluate model generalization. This approach guarantees equal class distribution in each fold, which minimizes the variations in performance estimates and enhances robustness against overfitting.

J. Best Model Selection

The final model was chosen due to the accuracy, cross validation results, ROC-AUC and the overall stability of the model. Random Forest and XGBoost models were compared and the one that showed best generalization and predictive capacity was selected for deployment.

K. Model Persistence

The final trained pipeline was saved using joblib to enable model reuse, deployment, and consistent inference. This ensures that preprocessing steps and model parameters remain identical during future predictions, supporting reproducibility in real-world applications.

The saved model can be integrated into mental health prediction systems for scalable and consistent deployment in production environments.

IV. Results and Discussions

This section presents the experimental evaluation, exploratory analysis, preprocessing, model performance, and interpretation of findings. The analysis follows the complete machine learning pipeline from raw data to final model selection.

A. Experimental Setup

Pre-processing resulted in a cleaned and deduplicated dataset with 97,723 records and 15 predictive features being used for the study. The dataset contains a variety of behavioral, psychological, occupational, and demographic variables, and can be used to train supervised classifiers.

Two ensemble learning models were implemented and evaluated under identical conditions:

- Random Forest Classifier
- XGBoost Classifier

An 80/20 stratified train-test split was applied to preserve class distribution across training and testing datasets.

Model evaluation was conducted using multiple performance metrics including:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix
- Cross-validation

This multi-metric evaluation provides a reliable assessment of model robustness and predictive capability for mental health prediction.

B. Read In and Exploration of Data

Initial exploration showed that the dataset was nearly balanced:

- 48.5% required mental health treatment
- 51.5% did not require treatment

This balanced distribution reduces classification bias and removes the need for resampling techniques such as SMOTE.

Key observations during exploration include:

- All variables are categorical in nature
- Behavioral and psychological variables dominate the feature space
- Only one feature *self_employed* contained missing values

The dataset structure was therefore highly suitable for tree-based ensemble learning methods.

C. Data Cleaning and Preprocessing

Several preprocessing operations were performed before model development:

- Missing values in *self_employed* were replaced with "Unknown"
- The *Timestamp* feature was removed due to irrelevance
- 194,641 duplicate records were removed

- Spark DataFrame was converted into Pandas format for visualization and machine learning
- Categorical variables were encoded using OrdinalEncoder

Random Forest and XGBoost are tree-based model that are not sensitive to the feature scale, so no feature scaling was necessary.

D. Data Analysis Findings

Exploratory analysis revealed several important psychological and behavioral patterns associated with treatment-seeking behavior.

1. Stress as a Key Indicator: Stress was a strong psychological risk indicator, with higher stress levels being associated with a higher likelihood of treatment.

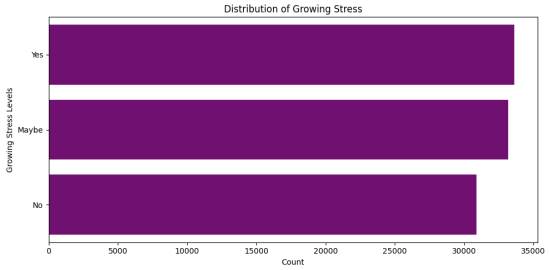


Fig. 6. Distribution of self-reported stress levels

As shown in Figure 6, stress is a major component in the dataset, as most people reported moderate to high stress levels.

2. Social Isolation Effect: People who had high levels of environmental isolation also expressed higher stress levels, indicating that environmental isolation is associated with psychological distress.

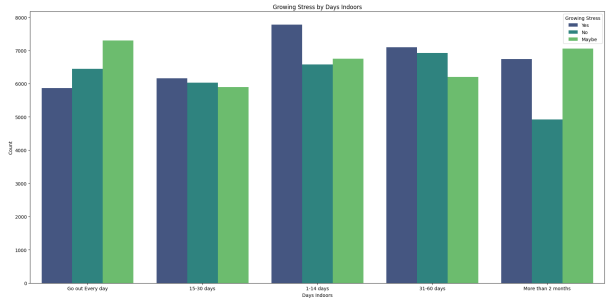


Fig. 7. Relationship between time spent indoors and stress levels

The figure shows that there is an apparent rise in stress among people who spend long periods of time indoors, as shown in Figure 7.

3. Family History as a Strong Predictor: One of the most robust predictors of treatment need was family history of mental illness.

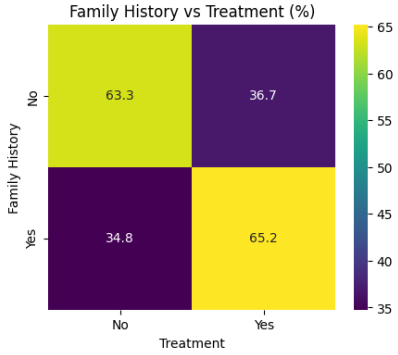


Fig. 8. Relationship between family history and treatment-seeking behavior

As seen in Figure 8, people with a family history are much more likely to seek mental health treatment.

4. Mood Instability as a Behavioral Signal: Mood swings showed a strong positive relationship with treatment likelihood.

E. Data Visualization Summary

Additional visualization analysis showed that females demonstrated higher treatment-seeking behavior compared to males, while occupational groups showed relatively similar treatment rates.

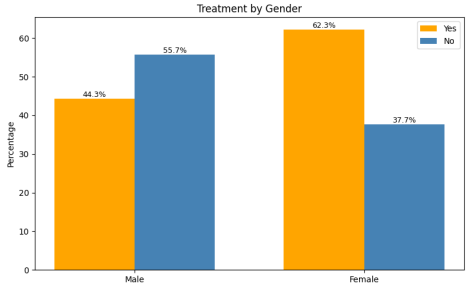


Fig. 9. Gender differences in treatment-seeking behavior

Figure 9 highlights that females reported higher treatment uptake than males, potentially reflecting greater awareness and openness toward mental health support.

F. Model Performance Results

The same experimental conditions were used for both models for comparison.

TABLE I
MODEL PERFORMANCE COMPARISON

Metric	Random Forest	XGBoost
Accuracy	0.771	0.814
Precision	0.775	0.818
Recall	0.771	0.814
F1-Score	0.771	0.814
Training Time (s)	574.37	353.13

XGBoost outperformed RF on all evaluation metrics, particularly for recall and overall classification accuracy.

G. Model Evaluation and Comparison

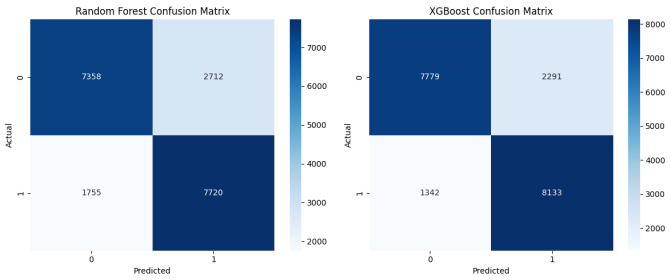


Fig. 10. Confusion matrix comparison between Random Forest and XGBoost

In Figure 10, we can see that XGBoost achieves much lower false negative rates than Random Forest and hence improves the early detection of people who need mental health treatment.

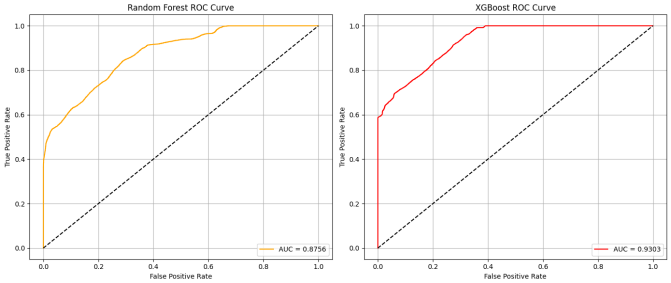


Fig. 11. ROC curve comparison between Random Forest and XGBoost

In Figure 11, the ROC analysis reveals that the model XGBoost obtained the highest score of ROC-AUC around 0.93, which indicates better class separation and more robust prediction ability.

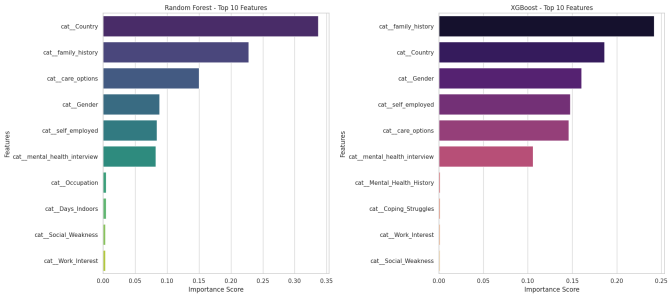


Fig. 12. Feature importance comparison between Random Forest and XGBoost

Family history, care options, growing stress and mood swings were the most significant features in both models, as determined by feature importance analysis in Figure 12.

H. Cross-Validation Results

TABLE II
CROSS-VALIDATION PERFORMANCE COMPARISON

Model	Mean Accuracy	Std Deviation
Random Forest	0.772	0.0025
XGBoost	0.812	0.0014

The cross-validation results also show that XGBoost not only has a higher predictive accuracy across the folds but also has a lower variance, indicating that it has a stronger generalization ability, and is more stable.

I. Discussion of Findings

The findings demonstrate that mental health treatment prediction depends on a combination of:

- Stable risk factors such as family history
- Behavioral indicators including stress and mood swings
- Environmental factors such as prolonged indoor isolation
- Psychological help-seeking behavior

This confirms that mental health risk is multi-dimensional and highly nonlinear, making ensemble learning methods particularly effective for prediction tasks involving complex behavioral data.

J. Analysis of Findings

From a machine learning perspective, XGBoost outperformed Random Forest due to:

- Sequential error correction through boosting
- Better optimization of difficult samples
- Strong regularization mechanisms
- Improved modeling of complex feature interactions

From a healthcare perspective, reducing false negatives is critically important because failing to identify high-risk individuals may delay psychological intervention and treatment support.

The results also validate the use of behavioral factors (stress and mood swings) as early warning signs and family history as a long-term structural risk factor.

K. Limitations and Ethical Considerations

The models were found to be performing well in predicting the data, but there are still a few drawbacks.

The data used are self-reported survey responses and may suffer from reporting bias, reporting inconsistencies, and subjectivity in mental health assessments. In addition, the dataset might not be representative of the entire demographic population and may not be generalizable to other real-world settings.

Ethically, machine learning systems for mental health prediction need to be used judiciously since an incorrect prediction could have psychological and/or social impacts on the individual. Consequently, the proposed system cannot be used instead of professional clinical diagnosis, but rather as an aid to clinical diagnosis in early screening and intervention.

L. Final Interpretation

In general, the results of the study showed that machine learning models could be used to predict the treatment needs of mental health using structured behavioral and psychological data.

XGBoost performed the best overall of all the models evaluated because:

- Highest classification accuracy (81.4%)
- Strong ROC-AUC score (0.93)
- Lower false negative rate
- Better generalization performance

These findings highlight the strong potential of machine learning systems as effective decision-support tools for early mental health screening when used ethically alongside professional clinical assessment.

V. Conclusion

The current study was able to successfully build scalable machine learning models to predict mental health treatment needs from large-scale behavioral, psychological, occupational, and demographic survey data. Apache Spark was used to efficiently deal with big data processing and handling, while Scikit-learn and XGBoost were used to develop and evaluate machine learning models. Experimental analysis showed that mental health treatment needs were significantly correlated with factors related to behavior and psychology including increased stress, mood swings, family history of mental illness, personal history of mental illness, and time spent indoors. These together were confirmed to be significant in identifying individuals at risk by exploratory analysis, visualization, preprocessing, and predictive modeling. Random Forest was outperformed by XGBoost on most of the evaluation metrics among the evaluated models, with an accuracy of 81.4% and a ROC-AUC score of 0.93. The model also showed superior generalization performance and lower false negative rates, especially crucial for predicting mental health as misidentification of high-risk individuals could delay early intervention and support.

While good predictive results were obtained, there are some drawbacks. Data presented was self-reported survey data and may have included subjectivity and reporting bias. Moreover, the study assessed solely ensemble learning algorithms and might not be applicable to other demographic groups and clinical healthcare settings. Future research could involve applying deep learning methods, EAI strategies, and real-time monitoring of mental health to enhance the predictive performance, explainability, and utility of the models. In conclusion, the results indicate that ethical machine learning systems can be successfully used in conjunction with clinical evaluation and healthcare knowledge to support decisions for early mental health risk screening, in a scalable manner.

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